1. Business Understanding

BC Analytics aims to improve early detection and management of diabetes risk. Current healthcare systems are largely reactive and lack tools to support proactive, data-driven decisions. This project addresses that gap by developing a decision support system that classifies patient diabetes risk, identifies key lifestyle drivers, and segments patients into meaningful groups for targeted interventions. The final solution includes an interactive dashboard to assist healthcare providers during consultations.

2. Data Understanding

The project uses the Diabetes_and_Lifestyle_Dataset, which contains patient health indicators and lifestyle variables. The target variable is diabetes_stage, representing different levels or types of diabetes risk. Initial exploration focused on understanding the distribution of diabetes stages and relationships between features such as BMI, physical activity, diet, and glucose levels. Observations indicated class imbalance and strong correlations between lifestyle behaviours and diabetes progression.

3. Data Preparation

Data preprocessing involved handling missing values, encoding categorical variables, and scaling numerical features where necessary. Outliers were assessed and treated to reduce model bias. Feature selection was guided by both domain knowledge and exploratory analysis, ensuring that relevant health and lifestyle indicators were retained. The dataset was then split into training and testing sets for model development.

4. Modelling

4.1 Risk Classification

Three models were implemented to predict diabetes_stage:

- Decision Tree
- Random Forest
- XGBoost

Random Forest and XGBoost outperformed the Decision Tree due to their ability to handle complex feature interactions and reduce overfitting. XGBoost achieved the best overall performance and was selected as the final model.

4.2 Patient Segmentation

K-means clustering (k=3) was applied to group patients based on lifestyle and health characteristics. The clusters revealed distinct patient profiles:

- Low-risk, healthy lifestyle group
- Moderate-risk group with mixed habits
- High-risk group with poor lifestyle indicators

These segments support targeted intervention strategies.

4.3 Key Driver Analysis

SHAP (Shapley Additive Explanations) was used to interpret both classification and clustering models. Key drivers of diabetes risk included:

- Glucose levels
- BMI
- Physical activity
- Dietary habits

SHAP values provided both global and individual-level explanations, making the model outputs interpretable for clinical use.

5. Evaluation

The models were evaluated using metrics such as accuracy, precision, recall, and F1-score. Out of the models tested, XGBoost performed the best overall, showing strong results on the test data.

There were a few misclassifications between similar diabetes stages, which is expected since some patients have very similar characteristics. However, the model was still able to separate the different risk levels quite well. For segmentation, the K-means model (k=3) created clear groups of patients. These groups made sense and reflected low-risk, moderate-risk, and high-risk profiles based on their lifestyle and health data.

SHAP values were used to understand how the model makes predictions. The most important features included glucose levels, BMI, physical activity, and diet, which are all well-known factors linked to diabetes risk. This helps show that the model is producing reasonable and reliable results.

Model Performance Summary

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	Good	Good	Good	Good
Random Forest	Better	Better	Better	Better

Model	Accuracy	Precision	Recall	F1-Score
XGBoost (Final)	Best	Best	Best	Best

6. Deployment

To make the system usable in practice, an interactive web application was built using Dash. The app allows users to enter patient details across three sections: demographics, lifestyle factors, and medical measurements.

After entering the data, the system predicts the patient's diabetes risk category in real time. It also shows the main factors influencing the prediction using a SHAP visualisation, which helps explain why a certain result was given.

The layout of the app is simple and easy to use, making it suitable for use during consultations. By combining predictions, explanations, and patient insights in one place, the application supports more informed and data-driven decision-making.

7. Conclusion & Recommendations

This project developed a decision support system to help identify diabetes risk and group patients based on their lifestyle and health data. XGBoost gave the best results for predicting diabetes stages, while K-means clustering helped group patients into meaningful categories.

The use of SHAP made the model easier to understand by showing which factors influence the predictions the most. Key factors included glucose levels, BMI, physical activity, and diet.

Based on the results, the following recommendations are proposed:

- Focus on early intervention for patients in the moderate-risk group
- Encourage lifestyle changes such as better diet and increased physical activity for high-risk patients
- Use the dashboard during consultations to support decision-making
- Regularly monitor important indicators like glucose levels and BMI

Overall, the system helps move towards a more proactive approach to healthcare by identifying risk earlier and supporting better patient management.